

Prostate-specific Antigen (PSA), Free:Total Ratio Reflex

Test Details

METHODOLOGY

Electrochemiluminescence immunoassay (ECLIA)

RESULT TURNAROUND TIME

Within 1 day

Turnaround time is defined as the usual number of days from the date of pickup of a specimen for testing to when the result is released to the ordering provider. In some cases, additional time should be allowed for additional confirmatory or additional reflex tests. Testing schedules may vary.

Test Includes

Total PSA; free PSA when total is between 4.0 and 10.0 ng/mL. If reflex testing is performed, additional charges/CPT code(s) may apply.

Use

Measure the percentage of free (uncomplexed) PSA relative to the total amount of PSA in men with serum PSA concentrations between 4.0 and 10.0 ng/mL.

Special Instructions

Specimens should not be drawn immediately after digital rectal examination (DRE), prostatic massage, or transrectal ultrasound (TRUS). PSA sampling should not be performed for at least six weeks after prostatic biopsy.

Values obtained with different assay methods should not be used interchangeably in serial testing. It is recommended that only one assay method be used consistently to monitor each patient's course of therapy. This procedure does not provide serial monitoring. It is intended for one-time use only. If serial monitoring is required, please use the serial monitoring number **480640** (/tests/480640/prostate-specific-antigen-psa-free-total-ratio-reflex-serial-monitor) to order.

Limitations

The interpretive guidelines provided for percent free PSA are based on a population of men with normal digital rectal exam (DRE) and total PSA between 4.0 and 10.0 ng/mL. Catalona and coworkers² did not make specific recommendations regarding the use of percent free PSA for any other population of men.

Patients taking finasteride, an alpha-reductase inhibitor, will have diminished levels of PSA. PSA complexes are more stable than free PSA.¹ The serum levels of both total and free PSA increase with prostate manipulation but the free returns to premanipulation concentrations quicker. This can result in a transient elevation in percent free PSA.

Custom Additional Information

In general, serum PSA levels increase due to physical changes to prostate architecture caused by trauma, infection, inflammation, prostate manipulation, benign prostatic hypertrophy (BPH), or malignancy.^{3,4} The sensitivity of PSA levels to these changes serves as the basis for the clinical use of the test. The PSA concentration in the serum of healthy men is a millionfold lower than that in seminal fluid. PSA in seminal fluid is predominantly free or uncomplexed. In serum, the majority of PSA is bound to inhibitors, including α_1 -antichymotrypsin (ACT) and α_2 -macroglobulin (A2M). Measured total PSA consists of free and ACT-bound, since PSA complexed to A2M is not immunologically detectable.

Catalona and coworkers found that one in four patients with normal DRE and PSA levels between 4.0 and 10.0 ng/mL have prostate cancer.² They recommend using a cutoff of 25% free PSA for this group of men (see Limitations) to identify individuals with an increased risk of prostate cancer. They found that 95% of men with cancer (as determined by biopsy) with normal DRE and total PSA between 4.0 and 10.0 ng/mL had percent free PSA of $\leq 25\%$. Their study further indicated that 20% of men with benign disease (as determined by biopsy) with normal DRE and a total PSA between 4.0 and 10.0 ng/mL had percent free PSA greater than the 25% cutoff.

Alternatively, percent free PSA may be used to determine the relative risk of prostate cancer in individual men.² The following table lists the probability of prostate cancer for men with nonsuspicious DRE results and total PSA between 4.0 and 10.0 ng/mL, by patient age (see Limitations). See table.

% Free PSA	50 to 64 Years	65 to 75 Years
0.00–10.00	56%	55%
10.01–15.00	24%	35%
15.01–20.00	17%	23%
20.01–25.00	10%	20%
>25	5%	9%

Specimen Requirements

Specimen

Serum (preferred) **or** plasma

Volume

1 mL

Minimum Volume

0.7 mL (**Note:** This volume does **not** allow for repeat testing.)

Container

Red-top tube, gel-barrier tube, **or** green-top (lithium heparin) tube. Do **not** use oxalate, EDTA, or citrate plasma.

Collection Instructions

If a red-top tube or plasma is used, transfer separated serum or plasma to a plastic transport tube.

Stability Requirements

Temperature	Period
Room temperature	3 days
Refrigerated	3 days
Frozen	14 days
Freeze/thaw cycles	Stable x3

Storage Instructions

Maintain specimen at room temperature.¹

Causes for Rejection

EDTA plasma specimen

Footnotes

1. Woodrum D, French C, Shamel LB. Stability of free prostate-specific antigen in serum samples under a variety of sample collection and storage conditions. *Urology*. 1996 Dec; 48(6A Suppl):33-39. PubMed 8973697

(<http://www.ncbi.nlm.nih.gov/pubmed/8973697>)

2. Catalona WJ, Partin AW, Slawin KM, et al. Use of percentage of free prostate-specific antigen to enhance differentiation of prostate cancer from benign prostatic disease. A prospective multicenter clinical trial. *JAMA*. 1998 May 20; 279(19):1542-1547. PubMed 9605898

(<http://www.ncbi.nlm.nih.gov/pubmed/9605898>)

3. Sokoll LJ, Chan DW. Prostate-specific antigen. Its discovery and biochemical characteristics. *Urol Clin North Am*. 1997 May; 24(2): 253-259. PubMed 9126221 (<http://www.ncbi.nlm.nih.gov/pubmed/9126221>)

4. Polascik TJ, Oesterling JE, Partin AW. Prostate-specific antigen: A decade of discovery--what we have learned and where we are going. *J Urol*. 1999 Aug; 162(2):293-306. PubMed 10411025

(<http://www.ncbi.nlm.nih.gov/pubmed/10411025>)

